

Promoting workplace safety in research on workers' unsafe behaviours in Iran

Background/Objectives

Accidents and injuries have been a regular occurrence in the petrochemical industries, because of the particular conditions dependent on its nature, over the last decade. Major occupational accidents that occur in this industry frequently result in significant financial and social losses. In this context, workers' unsafe behaviours are the main causes of work-related accidents and injuries. The first step to promote safe behaviours and reduce accidents and losses in petrochemical industries is identification of the influencing factors associated with workers' unsafe behaviours. The aim of this study was to (a) explore participants' perceptions of workplace safety; and (b) to identify the factors that impede safe behaviours by workers.

Methods

A qualitative study was conducted and reported according to analysis steps as described by Graneheim and Lundman for qualitative research. Eighteen participants were recruited using purposive and snowball sampling techniques, from petrochemical industry in Iran. Individual face-to-face semi-structured interviews were conducted to gain in-depth understanding of factors acting as a barrier for workers' safe behaviours and transcribed in Persian and then translated into English. Conventional content analysis was performed.

Results

Main themes emerging from the interviews were: (i) poor direct safety management and supervision; (ii) unsafe workplace conditions; (iii) workers' perceptions, skills and training; and (iv) broader organisational factors. They give insights into the effective measures which managers can implement to improve safe behaviours.

Discussion

These measures can eliminate the risk of work accidents and improve safety in other parts of the world where these are issues for workers.

Keywords

Unsafe behaviours, Workplace safety, Petrochemical industry, Workers

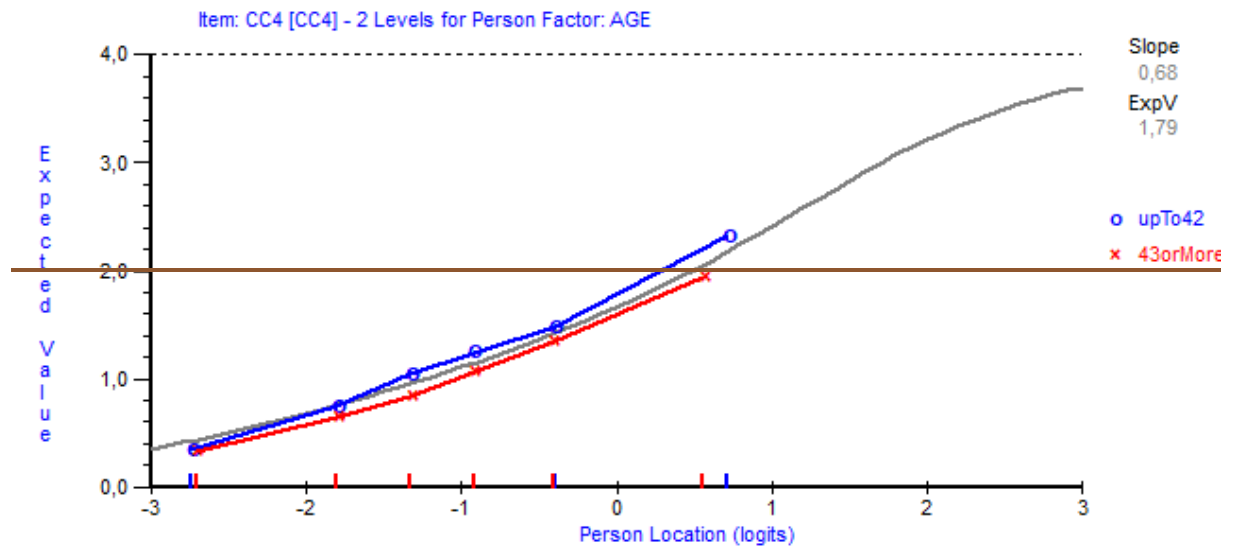


Fig 1. The item characteristic curve of item C14 (*When I'm working, I have trouble concentrating*) for younger (up to median age of 42) and older (over median age of 42) participants in a sample from Austria (n=1054).

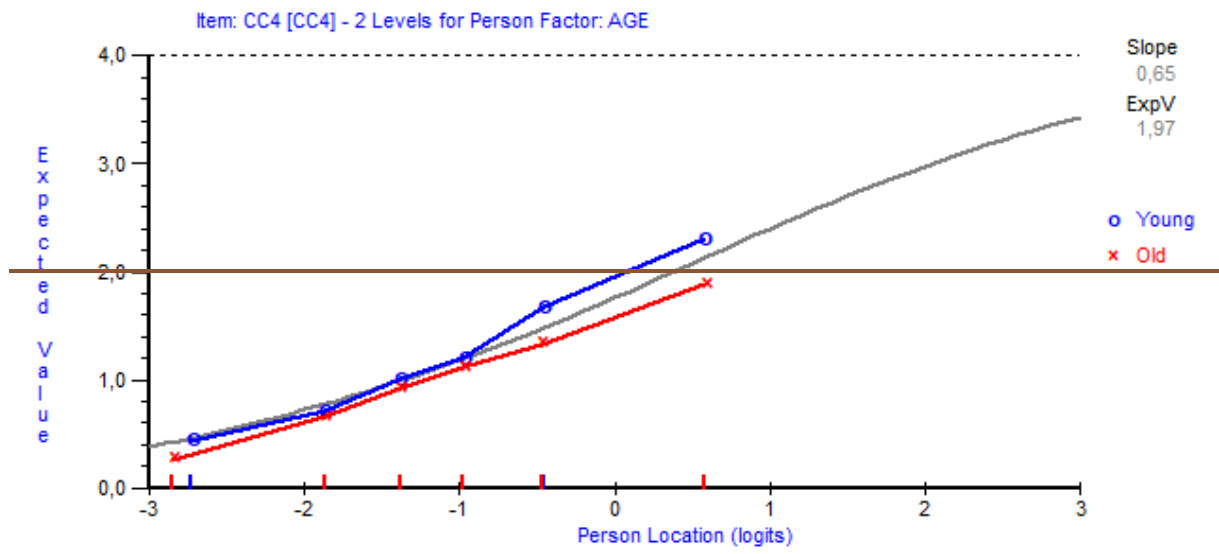


Fig 2. The item characteristic curve of item CI4 (*When I'm working, I have trouble concentrating*) for younger (up to median age of 42) and older (over median age of 42) participants in a sample from Germany (n=1073).

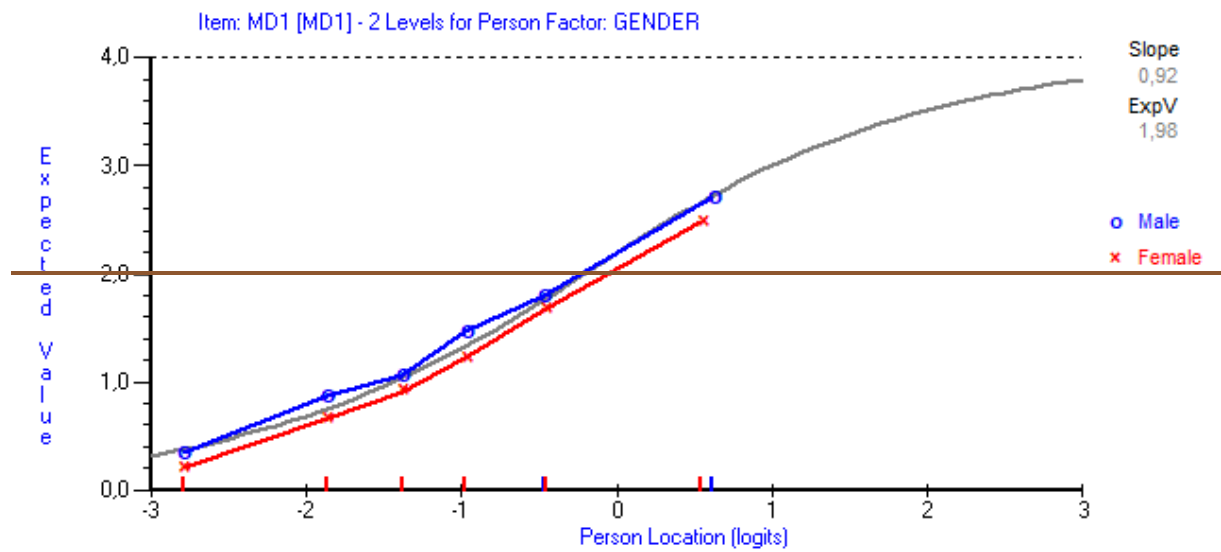


Fig 3. The item characteristic curve of item MD1 (I struggle to find any enthusiasm for my work) for women and men in a sample from Germany (n=1073).

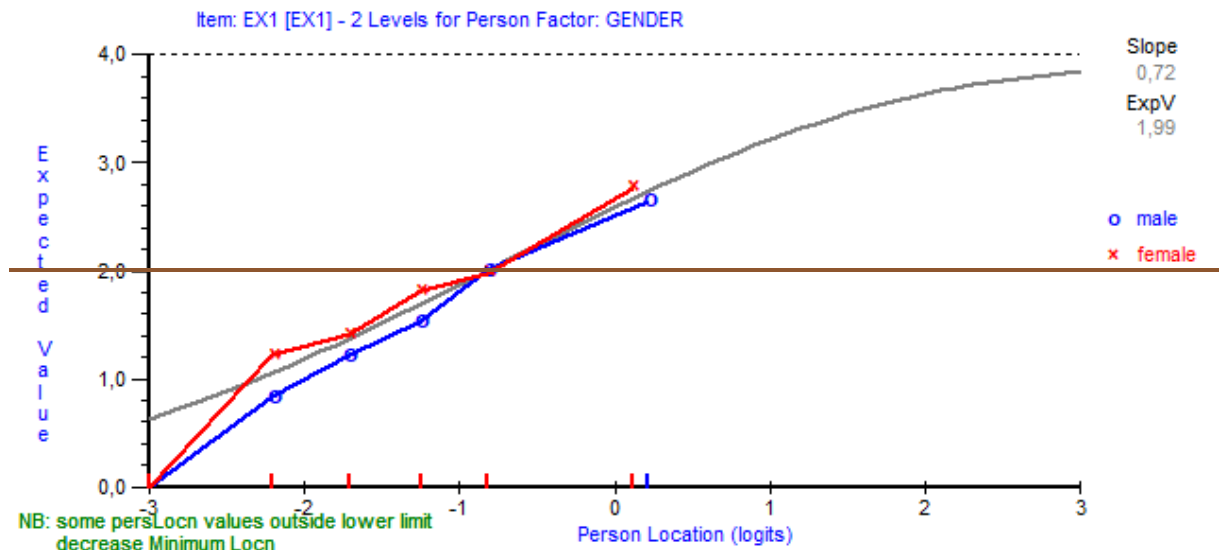


Fig 4. The item characteristic curve of item EX1 (*At work, I feel mentally exhausted*) for women and men in a sample from Check Republic (n=964).

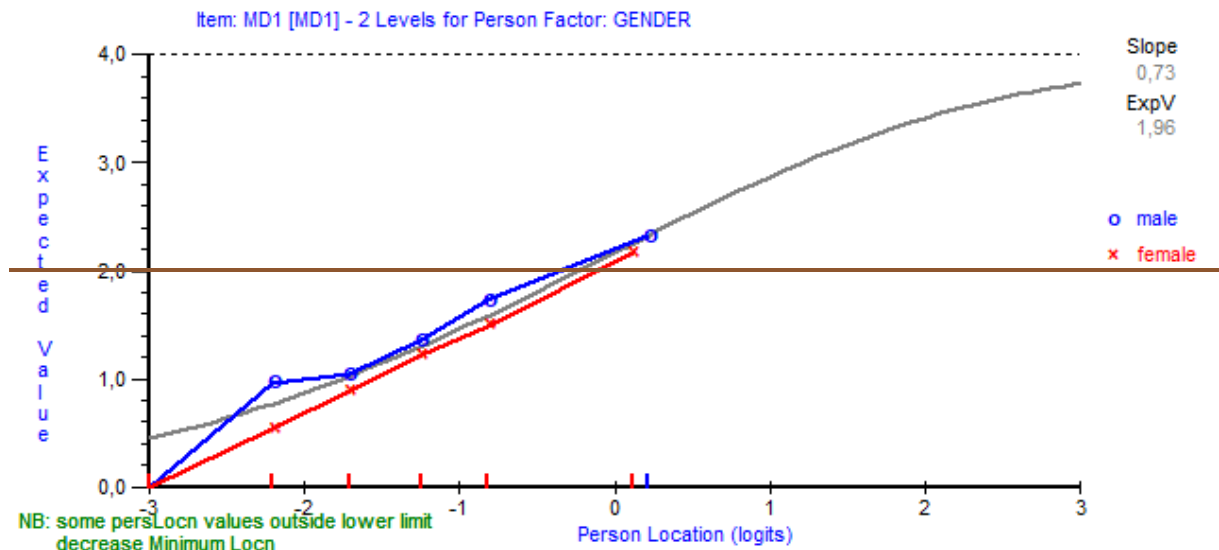


Fig 5. The item characteristic curve of item MD1 (I struggle to find any enthusiasm for my work) for women and men in a sample from Czech Republic (n=964).

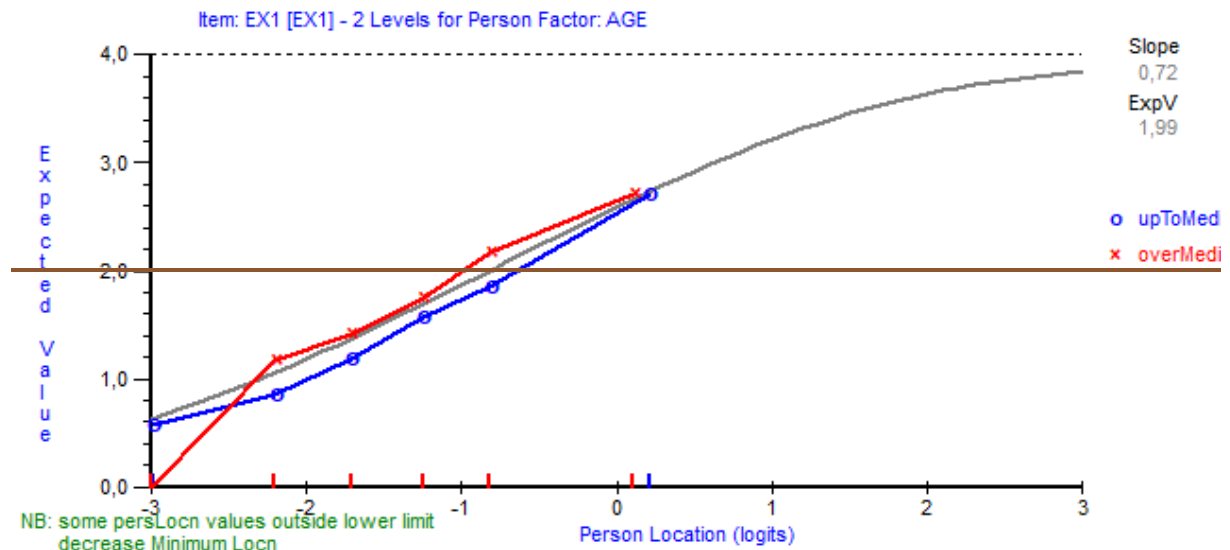


Fig 6. The item characteristic curve of item EX1 (*At work, I feel mentally exhausted*) for younger (up to median age of 42) and older (over median age of 42) participants in a sample from Czech Republic (n=964).

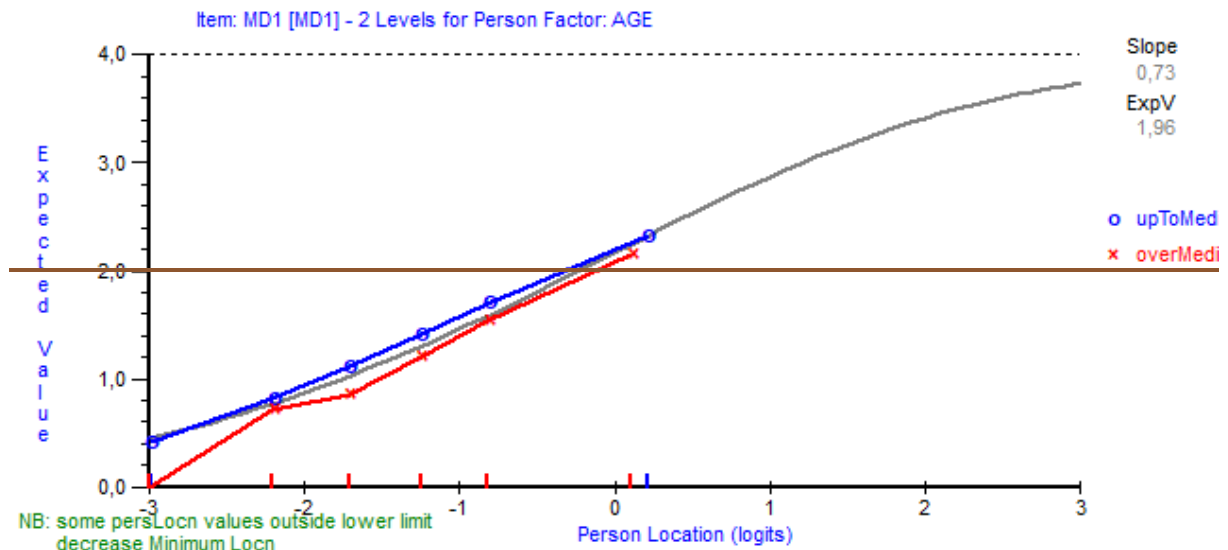


Fig 7. The item characteristic curve of item MD1 (I struggle to find any enthusiasm for my work) for younger (up to median age of 42) and older (over median age of 42) participants in a sample from Czech Republic (n=964).

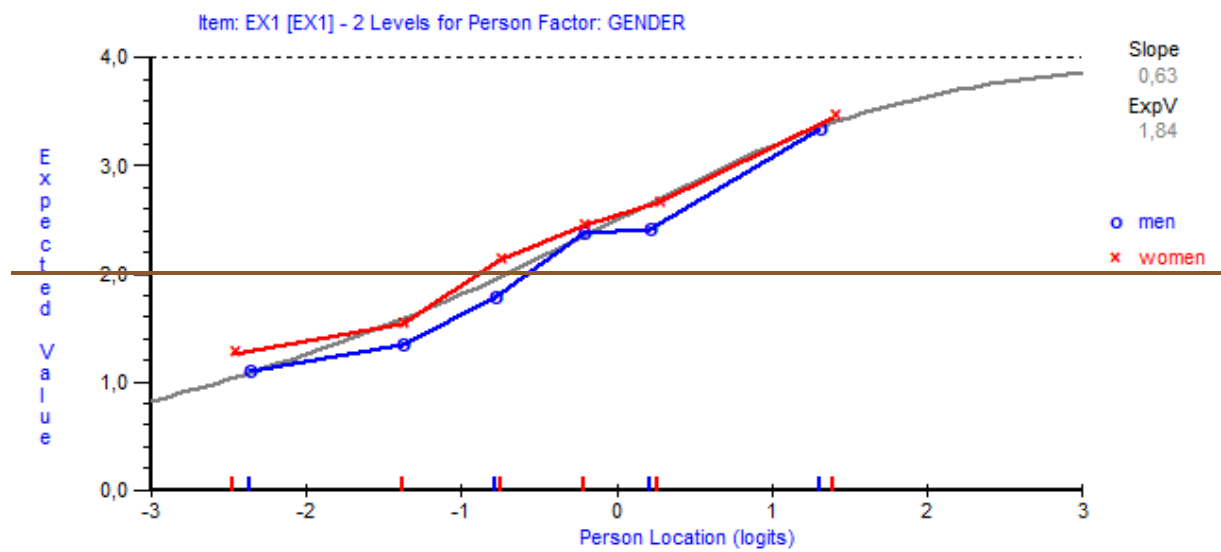


Fig 8. The item characteristic curve of item EX1 (*At work, I feel mentally exhausted*) for women and men in a sample from Japan (n=1028).

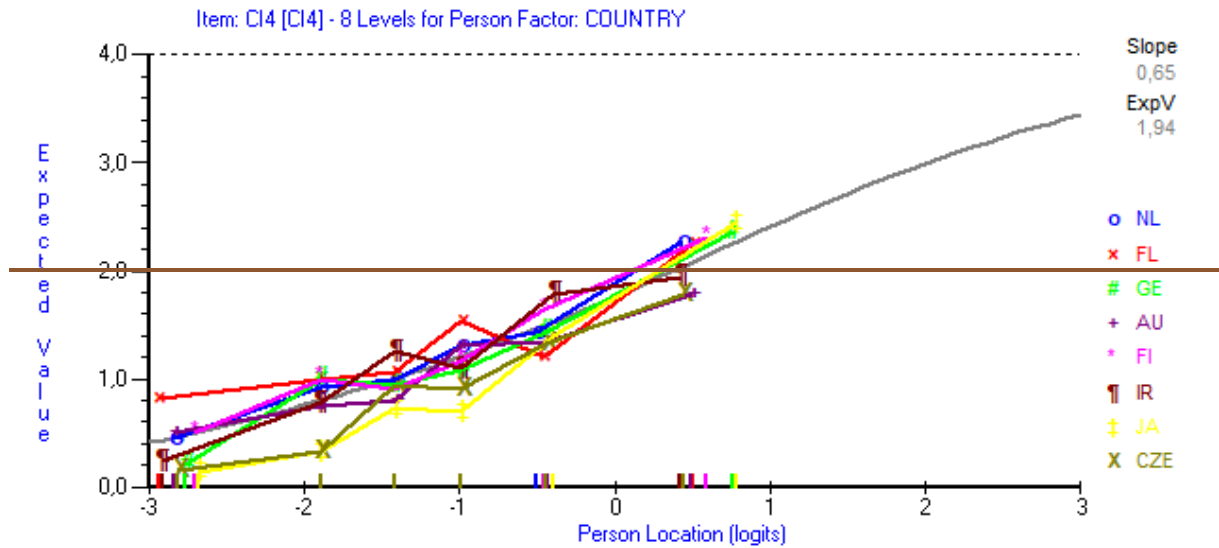


Fig-9. The item characteristic curve of item CI4 (*When I'm working, I have trouble concentrating*) for participants in different countries (NL= Netherlands, FL= Belgium, Flanders, GE=Germany, AU=Austria, FI=Finland, IR=Ireland, JA=Japan, CZE= Czech Republic) in a sample including 100 randomly selected participant from each country (n=800).

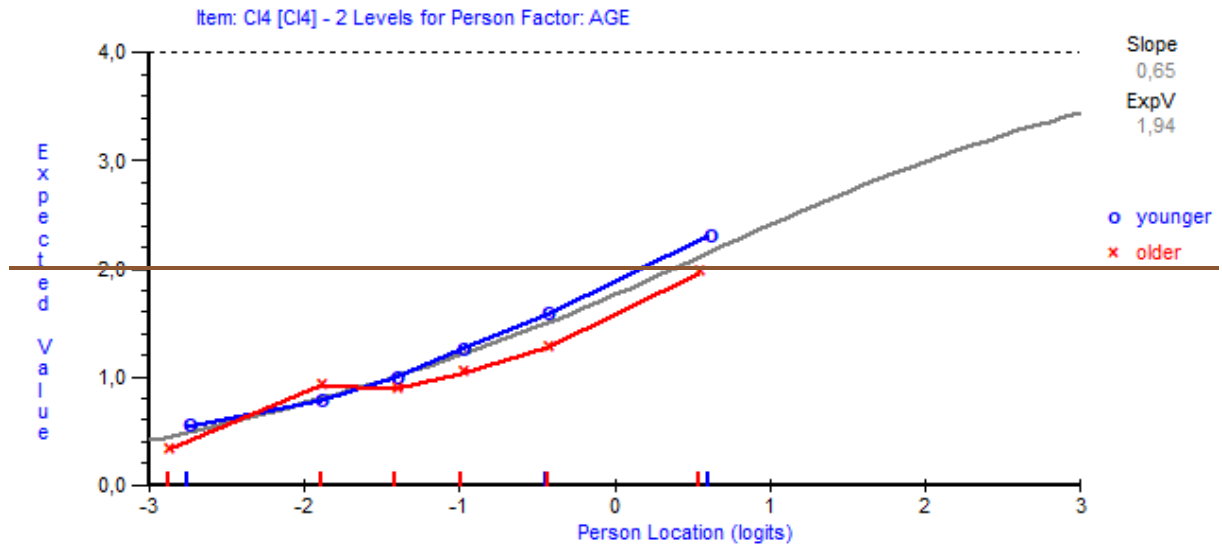


Fig-10. The item-characteristic curve of item CI4 (*When I'm working, I have trouble concentrating*) younger (under the median age) and older (above the median age) participants in a sample including 100 randomly selected participant from following countries Netherlands, Belgium, Flanders, Germany, Austria, Finland, Ireland, Japan, Czech Republic) (n=800).